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# Pattern Arithmetic: Phase-Colored Geometric Framework for High-Density Information via Unit Probability Spheres

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## Abstract

Ordinary arithmetic treats the integer 1 as a unit of truth:  $1 = 1$  and  $1 + 1 = 2$ . Counting is only one algebra. Colors and musical notes are also units of truth; combining them yields a hue or a chord, not a larger integer. This paper treats a pattern as a higher-order unit and represents one pattern by one typed point  $x = (r, \hat{n}, \chi, \tau)$  on a unit sphere  $\mathcal{S} \subset \mathbb{R}^3$ : strength  $r \in [0, 1]$ , direction  $\hat{n} \in S^2$  (equivalently  $(\theta, \phi)$ ), internal phase-color  $\chi \in [0, 2\pi)$ , and sort  $\tau$ . Pattern Arithmetic is the algebra of such units. It proceeds in three acts — collapse (the bowl,  $\oplus$ ), transform (the path,  $T$ ), result — so that identical states do not double ( $A \oplus A = A$ ), a gathered unit may be removed ( $A \ominus A = \emptyset$ , black), and the same ingredients may form different ordered expressions. Anti-phase cancellation to black is a Mix reading, not a gather. Words and numbers occupy the same sphere under different encodings and meet in a document by order and typed action, not by blending. The aim is a single bounded register that carries overlapping identities, structural paths, and mixed prose without a discrete grid.

## 1 Introduction

Ordinary arithmetic uses the integer 1 as a unit of truth. The identity  $1 = 1$  says that two tokens are of the same kind. The rule  $1 + 1 = 2$  says that combining two such tokens produces a larger quantitative state. That algebra answers the question *how many?*

It is not the only algebra. A color and a musical note are also units of truth, but their combination does not enlarge a pile. Under additive light, red and green yield yellow. Two copies of the same yellow, once brightness is normalized, remain yellow. Two notes

yield a chord, not a larger integer; a tone plus its opposite phase yields silence. In each case the word “add” names a different rule, and therefore a different kind of unit.

When the object of interest is a structured whole rather than a count, a hue, or a pitch, the natural unit is a *pattern*. This paper represents one pattern by one typed point on a unit sphere  $\mathcal{S} \subset \mathbb{R}^3$ , not by a field painted over the globe:

- $r \in [0, 1]$  is the strength of the unit, capped so that a state remains a single unit of truth — an activation, not a global probability density;
- $\hat{n} \in S^2$ , or equivalently  $(\theta, \phi)$ , is the direction of the pattern on the sphere;
- $\chi \in [0, 2\pi)$  is an internal phase-color, so two identities may occupy the same direction without overwriting one another;
- $\tau$  is the sort (**word**, **num**, **color**, ...): a legality tag, not a fifth geometric angle.

Pattern Arithmetic is the algebra of these units. Identical states do not double ( $A \oplus A = A$ ). A gathered unit may be removed ( $A \ominus A = \emptyset$ ). Rotations of the sphere preserve the unit. Opposite-phase Mix is a later reading; it is not  $\oplus$ .

A single numerical picture is enough to see the difference from counting. Take two unit states at the same place, one with phase  $0^\circ$  (red) and one with phase  $120^\circ$  (green). Counting would return 2. Phase-aware combination returns a single unit whose blended phase is  $60^\circ$  (yellow). The same two strengths with opposite phase ( $0^\circ$  and  $180^\circ$ ) return the null state, black. Two units of truth become either one new identity or none.

The working order of the paper is not that mix alone. It is a three-step spine:

$$\text{Collapse (bowl)} \longrightarrow T \text{ (path)} \longrightarrow \text{Result}.$$

Collecting red and green is not yet yellow, and not yet a poem. Yellow is what you get if the later path is  $T_{\text{paint}}$ . Red and green may equally well walk a  $T_{\text{poem}}$  path; the result need not be yellow — though a given poem may still arrive there. The pathway, not the shells, chooses the result. A verse is another path. Each word of “Twinkle, twinkle, little star...” is a unit state on the same sphere. Collapse yields the bag of distinct words (**twinkle** appears twice in the line and only once in the bowl). The poem is the path that walks those states in order, including the repeated **twinkle**. The bag is not the song. Numericity is not stored by  $\oplus$ . Two identical units collapse to one address; that is the point of  $A \oplus A = A$ . How many times the address is *visited* lives in the path (the tape says **twinkle** twice). How many as a *count* lives in **num** mode, where  $1 + 1 + 1 + 1 = 4$  and four tokens of 1 are not collapsed. Strength  $r$  is a poor substitute for either: a second **twinkle** cannot honestly raise  $r$  past 1, and “four ones” is schoolbook addition, not a fuller unit of truth. The same bowl may be read more than one way. Loading is not meaning.

The sphere is the stage. It is not the deepest claim. Strip away geometry, phase, rotations, and the language examples, and one proposition remains: different kinds of information require different notions of combination, and an operation should be read by what informational property it *preserves* — and what it is allowed to throw away.

$$1 + 1 = 2, \quad \text{Mix}(\text{red}, \text{green}) = \text{yellow}, \quad A \oplus A = A, \quad T_a(T_b(x)) \neq T_b(T_a(x)).$$

These are not alternative tricks for the same plus sign. They answer different questions:

| Operation            | Question                            |
|----------------------|-------------------------------------|
| +                    | How many?                           |
| $\oplus$ / $\ominus$ | What is present? / what is removed? |
| Mix                  | What composite state results?       |
| $T$                  | What happens, and in what order?    |
| offset               | What walk is reused?                |

A pattern can be treated as one unit. Identity and multiplicity come apart:  $\oplus$  keeps kind and discards count; the path keeps visits; **num** keeps quantity. Composition and transformation are different verbs. Sorts may share a stage without sharing an operator. The geometry that follows is only a language in which those verbs can be written. It does not mint the relations; a lexicon or a learning rule must still seat man and king.

## 1.1 The Limitations of Discretized Matrix Grids

A discrete matrix grid is the wrong box for such a unit. Traditional matrix representations of continuous state spaces suffer from two severe structural bottlenecks:

- **Memory Scaling Bottlenecks:** A voxel cube of resolution  $N$  stores  $O(N^3)$  cells, growing steeply with resolution.
- **Signal Identity Destruction:** Mapping multiple overlapping variables to identical spatial coordinates often results in the destructive overwriting of underlying signal identities.

## 1.2 The Phase-Colored Unit Sphere Framework

To resolve these grid constraints, the sphere is offered as one bounded object that can carry overlapping identities by direction and phase, without enlarging a lattice.

- **Spatial Orientation**  $\hat{n} \in S^2$ , or  $(\theta, \phi)$ : Defines continuous directional coordinates across the spherical surface.
- **Magnitude** ( $0 \leq r \leq 1$ ): Strength of that one unit, capped so the arrow remains a single unit of truth. An activation, not a density for the whole sphere.
- **Internal Color Phase** ( $\chi \in [0, 2\pi)$ ): Serves as an independent phase degree of freedom.
- **Sort**  $\tau$ : A legality tag (**word**, **num**, **color**, ...). It does not move the arrow.

## 1.3 Core Contributions & High-Level Intuition

The principal contributions of Pattern Arithmetic are:

- **A typed unit:** one atom  $x = (r, \hat{n}, \chi, \tau)$  on a bounded sphere, not a voxel cell and not a global probability field.
- **Three verbs on that unit:**  $\oplus$  collects identities ( $A \oplus A = A$ );  $T$  walks an ordered path; offset copies a relation. Schoolbook  $+$  is reserved for **num**.

Later sections distinguish a numeric encoding from a lexical encoding on the same sphere, so that a report may carry words and numbers together without forcing them through  $\oplus$ .

The next section turns these definitions into formal mathematical axioms.

## 2 Axiomatic Foundations Of Pattern Arithmetic

### 2.1 Process principle: Collapse, path, result

Pattern Arithmetic proceeds in three named acts, kept separate so that a bowl may grow before it is read:

$$\underbrace{A_1 \oplus A_2 \oplus \cdots \oplus A_n}_{R = \text{the bowl}} \longrightarrow T_{w_k} \circ \cdots \circ T_{w_1}(R) \longrightarrow \text{Result}. \quad (1)$$

Here  $\oplus$  *collects*: identical tokens do not double ( $A \oplus A = A$ ), and opposite marks may sit together until a later reading.  $T$  is an ordered path, not a mix. The result is whatever that path (or another named reading) makes of  $R$  — a poem, a report, a staged endpoint.

Paint is one species of path, written  $T_{\text{paint}}$ :

$$T_{\text{paint}}(\{\text{red}, \text{green}\}) = \text{yellow}.$$

The same pair may take  $T_{\text{poem}}$  instead. That result is a verse about the two colors, not a hue — unless the poem itself walks back to yellow. Shells do not own a pathway. A verse is another species. One bowl, several pathways. One may collect and read in a single gesture; the algebra does not force the merge. This principle is an axiom of *order of acts*, not a new formula for a 4-tuple.

### 2.2 Axiom 1: The Normalized Unit Truth State ( $r \leq 1$ )

A pattern is one typed point, not a field painted over the globe. The working atom is

$$x = (r, \hat{n}, \chi, \tau), \quad \begin{array}{ll} r \in [0, 1], & \hat{n} \in S^2, \\ \chi \in [0, 2\pi), & \tau \in \{\text{word}, \text{num}, \text{color}, \dots\}. \end{array} \quad (2)$$

Equivalently,  $\hat{n}$  may be written in polar angles  $(\theta, \phi)$ , so the geometric part of the atom is the 4-tuple  $(r, \theta, \phi, \chi)$  used by the operators below. The sort  $\tau$  does not move the arrow; it says which buttons may be pressed (Section 5). A collection (bowl) is a separate finite set  $B = \{x_1, \dots, x_n\}$  of such atoms. The sphere holds one current occupant of the register; the bowl is who is present.

The cap  $r \leq 1$  applies to *that* arrow. It is a normalized strength / activation coordinate, not a global probability density and not a demand that all arrows on  $\mathcal{S}$  integrate to one. Two states may occupy different directions at once; each still carries its own strength  $r \leq 1$ . The sphere is the stage. The unit is the vector.  $r = 1$  is a full-strength mark. Nothing in this axiom sums the globe into a single global budget. A field of many atoms, if wanted later, is a different object (for example a superposition of Dirac masses at those points), not the atom itself.

## 2.3 Axiom 2: The Absolute Null State ( $r = 0$ , Black)

The lower bound of a single unit is vanishing strength:

$$r = 0 \implies x = \mathbf{0} \text{ (Black)}. \quad (3)$$

Direction and phase are then unused: there is no arrow to point. Black is an empty seat, not a field that is zero at every angle.

## 2.4 Axiom 3: Saturation (White)

White is the upper pole of the same bounded stage. It is not an infinite norm. Two readings are allowed; neither leaves the cap  $r \leq 1$ .

**White as a colour.** A single arrow may take a full-spectrum name (white light as one identity). That state is still one 4-tuple with  $r \leq 1$ .

**White as a full house.** The bowl may be saturated: many directions occupied, many hues present. Collecting further distinct shells no longer enlarges a unit past  $r = 1$ . The stage looks white because the palette is on. The catalogue of names is large — on the order of a 24-bit palette,  $\sim 16$  million colours per direction — and may be called practically endless. No unit is  $r = \infty$ .

$$\text{White} = \text{saturation of the bowl, or a full-spectrum name.} \quad (4)$$

Black is empty. White is as full as this construction allows. The menu is huge. The arrows stay finite.

## 2.5 Axiom 4: Gather and remove ( $\oplus$ , $\ominus$ )

$\oplus$  *collects* a unit into the bowl. It does not mix colours and it does not turn an arrow. Identical names occupy one seat:

$$A \oplus A = A. \quad (5)$$

Opposite marks may sit together:  $\{+1\} \oplus \{-1\} = \{+1, -1\}$ . The bowl is a set of states, not a blended 4-tuple.

$\ominus$  *removes* a unit. If that unit was the last occupant, the seat is black (Axiom 2.3):

$$\{A\} \ominus A = \emptyset. \quad (6)$$

$\ominus$  undoes a fresh gather. It is not anti-phase interference. Idempotence then gives the counting-trap

$$A \oplus A \oplus A \oplus A \ominus A = \emptyset \text{ (black)}. \quad (7)$$

Four gathers of the same name are still one seat; one removal empties it. Schoolbook  $1 + 1 + 1 + 1 - 1 = 3$  lives only in **num** mode.

## 2.6 Axiom 5: Mix as a reading ( $T_{\text{paint}}$ )

A blended 4-tuple appears only when a path *reads* the bowl. Write  $\text{Mix} = T_{\text{paint}}$ . Same place first; then a first recipe when directions differ. In both cases the output is  $(r', \theta', \phi', \chi')$  with  $r' \leq 1$ , or black.

**Same direction.** Treat strength and phase as a phasor  $z = re^{i\chi}$ . Then

$$z' = r_A e^{i\chi_A} + r_B e^{i\chi_B}, \quad r' = \min(1, |z'|), \quad \chi' = \arg(z'), \quad (8)$$

and  $(\theta', \phi') = (\theta_A, \phi_A)$ . If  $z' = 0$ , the result is black. Equal unit paints at  $0^\circ$  and  $120^\circ$  give  $z' = e^{i\pi/3}$ , so

$$\text{Mix}(1\angle 0^\circ, 1\angle 120^\circ) = (1, \theta, \phi, 60^\circ) \quad (\text{yellow}).$$

The unequal pair  $r = 0.6$  at  $0^\circ$  and  $r = 0.8$  at  $120^\circ$  gives  $z' = 0.20 + 0.6928i$ , hence

$$r' \approx 0.721, \quad \chi' \approx 73.9^\circ.$$

Equal anti-phase marks, same place, give  $z' = 0$  (black).

**Different directions (first recipe).** Let  $\hat{n}$  be the unit direction of a state and  $\mathbf{s} = r_A \hat{n}_A + r_B \hat{n}_B$ . If  $\mathbf{s} = \mathbf{0}$ , return black when the phasors also cancel; otherwise

$$\hat{n}' = \mathbf{s}/\|\mathbf{s}\|, \quad r' = \min(1, \|\mathbf{s}\|),$$

and  $\chi'$  still from (8). This recipe is marked as such; it closes the map  $\text{Mix} : \mathcal{S} \times \mathcal{S} \rightarrow \mathcal{S}$ .

$\oplus$  does not run this rule. Two shells in a bowl have no resultant angle until Mix (or another path) is applied.

## 2.7 Axiom 6: Rotational Conservation ( $\times \mathbf{R}$ )

A rotation  $\mathbf{R} \in \text{SO}(3)$  turns the direction  $\hat{n}$  and leaves strength, phase, and sort alone:

$$\mathbf{R} \cdot (r, \hat{n}, \chi, \tau) = (r, \mathbf{R}\hat{n}, \chi, \tau), \quad \|\mathbf{R}\hat{n}\| = \|\hat{n}\| = 1. \quad (9)$$

The unit of truth is preserved because  $r$  is not scaled and  $\hat{n}$  stays on  $S^2$ . This is the geometric engine of  $T$ , not a second blend.

# 3 Transformation Algebra: From Identity to Expression

## 3.1 Motivation: Two Layers of Meaning

The operators established so far ( $\oplus$ ,  $\ominus$ ) answer the question *who is present?* Combining Sky, Sun, and Wind under  $\oplus$  yields one order-independent bowl — a bag of seats, not a blended hue. Paint is Mix, a later reading. Analogies such as ruler-man-king are *not* gathers; they are deferred to the offset operator of Section 4. Gather  $\oplus$  does not, and structurally cannot, distinguish a poem in which the sun rises through a wind-blown sky from a poem in which wind extinguishes a darkening sun — both share the identical bowl  $R = \text{Sky} \oplus \text{Sun} \oplus \text{Wind}$ , since  $\oplus$  is commutative and idempotent by Axiom 2.5.

The distinction between these two poems is not a difference in *ingredients*; it is a difference in *how one state acts upon another, and in what order*. This is a *relational* operation, and it requires a second operator,  $T$ , layered on top of  $\oplus$ : where  $\oplus$  builds nouns,  $T$  builds sentences. A poem is not the resultant  $R$  itself, but a specific trajectory traced across  $\mathcal{S}$  by successive transformations applied to  $R$ .

### 3.2 Formal Definition of the Transformation Operator

Rather than introduce  $T$  as an unrelated new primitive, it is defined using the same rotational machinery already given in Axiom 2.7: every state  $w \in \mathcal{S}$  acts on another state  $x \in \mathcal{S}$  by rotating  $x$ 's direction about  $w$ 's own axis, by an angle proportional to  $w$ 's strength, while additively shifting  $x$ 's phase by  $w$ 's phase:

$$T_w(x) = \left( r_x, \text{Rot}_{\hat{n}_w}(\alpha(r_w))\hat{n}_x, (\chi_x + \chi_w) \bmod 2\pi, \tau_x \right) \quad (10)$$

where  $\hat{n}_w$  is the unit axis of  $w$ ,  $\text{Rot}_{\hat{n}}(\alpha) \in \text{SO}(3)$  is the rotation of Axiom 2.7 about that axis by angle  $\alpha$ , and  $\alpha(r_w) = \pi r_w$  is the simplest monotonic choice mapping strength to rotation angle (a full-strength unit,  $r_w = 1$ , rotates by the maximal  $180^\circ$ ). Magnitude  $r_x$  and sort  $\tau_x$  are left invariant: a transformation reorients and recolors a state, it does not strengthen or weaken it, and it does not recast it — those remain the roles of  $\oplus$  and of an explicit cast.

### 3.3 Derived Properties

The following properties follow from Equation (10) rather than being separately postulated.

**Identity.** If  $w = \mathbf{0}$  (the Null State of Axiom 2.3), then  $\alpha(0) = 0$  and the phase shift is 0, so  $T_0 = \text{id}$ .

**Invertibility.** The inverse of  $T_w$  is not “the same axis and the same  $+\alpha$  again.” Same axis, same positive angle, repeats the turn. Inverse is  $-\alpha$  on the same axis, or  $+\alpha$  on the antipodal axis, together with  $-\chi_w$ . Define

$$w^* = (r_w, -\hat{n}_w, -\chi_w \bmod 2\pi, \tau_w), \quad (11)$$

where  $-\hat{n}_w$  is the antipode ( $\pi - \theta_w, \phi_w + \pi$ ). Then  $T_{w^*}(T_w(x)) = x$  for all  $x$ . As rotations, only  $\alpha = 0$  ( $r_w = 0$ ) and  $\alpha = 180^\circ$  ( $r_w = 1$ ) equal their own reverse; the inverse state  $w^*$  is still required so that phase is cancelled too. This undoes a transformation on a state that persists. It is distinct from  $A \ominus A = \emptyset$  (last shell removed) and from Mix of equal anti-phase marks (a reading into black). Neither of those is an inverse of  $T$ .

**Associativity.** Since each  $T_w$  is a function  $\mathcal{S} \rightarrow \mathcal{S}$ , composition of transformations is associative automatically, by the associativity of function composition; no additional axiom is required.

**Non-commutativity.** Rotations about distinct axes in  $\text{SO}(3)$  generically do not commute: for  $a, b$  with distinct directions  $(\theta, \phi)$ ,

$$T_a(T_b(x)) \neq T_b(T_a(x)). \quad (12)$$

This is not an additional assumption; it is inherited directly from the well-established non-commutativity of  $\text{SO}(3)$ , already invoked by Axiom 2.7. It is this property, and only this property, that allows two poems built from the same resultant  $R$  to diverge: order of application is meaningful even though the underlying identity is not.

### 3.4 Poems as Transformation Trajectories

A poem  $\pi$  built from resultant  $R$  and an ordered sequence of transforming states  $w_1, w_2, \dots, w_n$  is the trajectory

$$\pi = (R; w_1, \dots, w_n) : R \longrightarrow T_{w_1}(R) \longrightarrow T_{w_2}T_{w_1}(R) \longrightarrow \dots \longrightarrow T_{w_n} \dots T_{w_1}(R). \quad (13)$$

Two poems may share an identical resultant  $R$  (the same theme, the same “ingredients”) while tracing entirely different trajectories — this is the formal sense in which many poems arise from the same units of truth. Conversely, because rotation composition is many-to-one, two distinct trajectories may terminate at the same endpoint; a poem’s identity is therefore carried by its full path through  $\mathcal{S}$ , not by its final state alone.

### 3.5 Worked Numerical Example: Sun, Wind, and Sky

Let Sky be the state acted upon, with direction  $(\theta, \phi) = (45^\circ, 45^\circ)$ , i.e. Cartesian axis  $(0.500, 0.500, 0.707)$ . Let Sun define an axis along  $(\theta, \phi) = (90^\circ, 0^\circ)$ , i.e.  $(1, 0, 0)$ , with strength  $r_{\text{Sun}} = 1.0$ , giving rotation angle  $\alpha_{\text{Sun}} = 180^\circ$ . Let Wind define an axis along  $(\theta, \phi) = (90^\circ, 90^\circ)$ , i.e.  $(0, 1, 0)$ , with strength  $r_{\text{Wind}} = 0.7$ , giving  $\alpha_{\text{Wind}} = 126^\circ$ .

*Order 1 — Sun acts first, then Wind* (the sun rising through a wind-blown sky):

$$T_{\text{Wind}}(T_{\text{Sun}}(\text{Sky})) \approx (-0.866, -0.500, 0.011) \quad (14)$$

*Order 2 — Wind acts first, then Sun* (wind extinguishing a darkening sun):

$$T_{\text{Sun}}(T_{\text{Wind}}(\text{Sky})) \approx (0.278, -0.500, 0.820) \quad (15)$$

The two endpoints are clearly distinct, confirming Equation (12): identical primitives, identical resultant  $R$ , opposite structural readings. Notably, if  $\chi_{\text{Sun}} = 0^\circ$  and  $\chi_{\text{Wind}} = 90^\circ$ , the accumulated phase shift is  $270^\circ$  in *both* orderings, since phase addition is commutative. The two poems therefore share the same blended color/mood even as their spatial — structural — meaning diverges. This asymmetry (shared affect, divergent structure) is a direct and testable prediction of the model.

### 3.6 Scope

$T$  is deliberately left unconstrained beyond Equation (10): any sequence of any transforming states is a formally valid trajectory. Pattern Arithmetic in this form supplies the substrate on which expression is built — the combinatorial space of possible trajectories — and does not yet supply a grammar constraining which trajectories are coherent, well-formed, or meaningful to a human reader. Whether such constraints belong inside this framework (as further axioms on admissible sequences of  $T$ ) or outside it (as a learned or externally specified grammar layered on top) is left open.

## 4 Analogical Arithmetic: The Offset Operator

Offset is a third verb. It is not  $\oplus$  (no blend) and not  $T$  (no spin of an actor on a path). It copies a *walk* from one seat to another. The landing is a new direction on the same globe. A dictionary may later pin a name on that landing. The walk does not mint the name.

## 4.1 Recipe: logarithm, carry, exponential

Write directions as unit vectors  $\hat{n} \in S^2 \subset \mathbb{R}^3$ . The walk from  $p$  to  $q$  is the tangent vector

$$\theta = \arccos(\hat{p} \cdot \hat{q}), \quad \log_{\hat{p}}(\hat{q}) = \frac{\theta}{\sin \theta} (\hat{q} - (\hat{p} \cdot \hat{q}) \hat{p}) \quad (\theta \neq 0), \quad (16)$$

with  $\log_{\hat{p}}(\hat{p}) = \mathbf{0}$ . Its length is the geodesic angle. The reverse map, from a tangent vector  $v$  at  $p$ , is

$$\exp_{\hat{p}}(v) = \cos \|v\| \hat{p} + \sin \|v\| \frac{v}{\|v\|} \quad (v \neq \mathbf{0}), \quad (17)$$

and  $\exp_{\hat{p}}(\mathbf{0}) = \hat{p}$ . Given three seats  $m, k, w$  (source, source-image, new start), set

$$v = \log_{\hat{n}_m}(\hat{n}_k), \quad \hat{n}_* = \exp_{\hat{n}_w}(\text{transport}_{m \rightarrow w}(v)). \quad (18)$$

For a first recipe, if  $m$  and  $w$  are not far, transport may be omitted and  $v$  used at  $w$  as written. That case is crude and marked as such. The full recipe carries  $v$  along the short geodesic  $m \rightarrow w$  so the heading stays a ground heading.

Strength and colour are not invented by the walk. This paper takes

$$r_* = r_w, \quad \chi_* = (\chi_w + \chi_k - \chi_m) \bmod 2\pi. \quad (19)$$

The target keeps its  $r$ . The mood increment of the pair  $m \rightarrow k$  is reused. Nothing here uses  $\alpha = \pi r$  from  $T$ . Two seated units keep two strengths. If king has  $r = 0.9$  and queen has  $r = 0.8$ , both sit; there is no  $0.9 + 0.8 = 1.7$ . That sum is ambient  $\mathbb{R}^3$  arithmetic and leaves the ball.  $\oplus$  gathers the two seats. A later path may turn a dial or clip a mix. It does not get to smash two  $r$ 's into one illegal length as the price of seating them.

## 4.2 A check on one equator, not a definition

Place three arrows in the  $xy$ -plane:

$$\hat{n}_m = (1, 0, 0), \quad \hat{n}_k = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}, 0\right), \quad \hat{n}_w = (0, 1, 0)$$

( $0^\circ, 30^\circ, 90^\circ$ ). Then  $\log_{\hat{n}_m}(\hat{n}_k) = (0, \pi/6, 0)$ . Carrying that vector a quarter-turn about  $z$  gives  $(-\pi/6, 0, 0)$ . The exponential at  $\hat{n}_w$  lands at

$$\hat{n}_* = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right) \quad (120^\circ).$$

On this slice the walk *looks like*  $90^\circ + 30^\circ = 120^\circ$ . That sum is a protractor check. It is not the operator, and it is not a queen. If a lexicon writes the names man, king, woman on the first three seats, it may write queen on the landing. The geometry only promised a fourth pin.

## 4.3 Offset is not $T$

$T_w(x)$  spins an occupant about an axis. Offset plants a new seat from three seats. A path does not need (18). A dictionary act does not need (10). Do not encode the walk as a ghost actor whose sole job is to trigger  $T$ .

The popular ambient sum  $\hat{n}_k - \hat{n}_m + \hat{n}_w$  then normalize (Recipe 1) is recorded as a cheaper cousin. This paper uses Recipe 2 so the copied step stays on the skin of the ball.

## 5 Encodings, Sorts, and Mixed Documents

The unit sphere is a single bounded *register*. Numbers, words, and colors are not three different globes; they are three encodings of a token onto the same globe. A scientific paragraph already mixes those alphabets on one page. Pattern Arithmetic may do the same, provided each loaded state carries a *sort* and the operators respect that sort.

### 5.1 Two primary sorts

A state  $x \in \mathcal{S}$  is the typed atom of Axiom 2.2. What the geometric 4-tuple  $(r, \theta, \phi, \chi)$  means depends on the sort  $\tau$  and the encoding used to write it:

- **Numeric sort (num).** The state represents a scalar. One toy encoding — the only numeric encoding used in this paper — stores an integer  $n$  on a marked clock as  $\chi = 2\pi n/N$  (arithmetic modulo  $N$ ), at a fixed direction, with  $r = 1$ . Another, stereo-style, is recorded as an option: it would send  $x \rightarrow \pm\infty$  toward the poles so that unbounded reals still occupy the globe. It is not used below.
- **Lexical sort (word).** The state is a labeled point in a toy dictionary living on the same globe. Similarity is the angle between points (cosine similarity). A bag of words is formed by  $\oplus$ . A sentence or poem is formed by  $T$ .

Schoolbook addition lives only in **num** mode. Pattern blend lives only in **word** (or color) mode. The two equations may share a sphere. They may not share an operator.

### 5.2 Numeric mode: $7 + 5 = 12$ on the clock, and why $\oplus$ is not 12

Fix  $N = 24$  and the encoding

$$E_{\text{num}}(k) = (r = 1, \hat{n} = (1, 0, 0), \chi = 2\pi k/24, \tau = \text{num}), \quad k \in \mathbb{Z}/24\mathbb{Z}. \quad (20)$$

Decode is the inverse chart:  $k = N\chi/(2\pi)$ , an integer hour mark. Then 7 sits at  $\chi = 105^\circ$ , 5 at  $\chi = 75^\circ$ , and 12 at  $\chi = 180^\circ$ . Schoolbook  $+$  is decoded arithmetic:

$$7 + 5 = E_{\text{num}}^{-1}(E_{\text{num}}(7)) + E_{\text{num}}^{-1}(E_{\text{num}}(5)) = 12, \quad E_{\text{num}}(12) = (1, (1, 0, 0), 180^\circ, \text{num}). \quad (21)$$

That is  $7 + 5 = 12$  as a calculation on  $\mathcal{S}$ : decode, add in  $\mathbb{Z}$ , encode. The type table of Section 5.4 refuses  $\oplus$  on a (num, num) pair. If one nevertheless applies Mix (Equation (8)) to the same two geometric marks, the phasors are

$$z = e^{i7\pi/12} + e^{i5\pi/12} = 2\cos(\pi/12)e^{i\pi/2}, \quad (22)$$

so  $r' = \min(1, 2\cos 15^\circ) = 1$  and  $\chi' = 90^\circ$ . The clock would read that mark as hour 6, not hour 12. A bowl  $\{E_{\text{num}}(7), E_{\text{num}}(5)\}$  is two seats, still not 12. This single contrast is why the type table exists:  $+$  answers *how many*;  $\oplus$  and Mix do not.

### 5.3 A document is a sequence, not a blend

Words and numbers coexist in reports because a document is an ordered tape of tokens, not one mixed arrow. Write a paragraph as a trajectory of typed atoms

$$\pi = (x_1, \dots, x_m), \quad x_i = (r_i, \hat{n}_i, \chi_i, \tau_i), \quad \tau_i \in \{\text{num}, \text{word}\}, \quad (23)$$

exactly as a poem was a trajectory in Section 3.4. The sphere holds one current occupant of the register; the paragraph is the history of loads and actions.

They may meet in only three honest ways.

**Juxtaposition.** The default. Tokens sit in order and no algebra is applied between them. In “the sample reached 37.4°C at dawn,” the number 37.4 is not blended with the word dawn.

**Typed action.** The operator  $T_w(x)$  is allowed across sorts only with an explicit output sort. Two useful conventions, which the rest of this paper adopts, are:

- a word acting on a number yields a number-in-context (still **num**);
- a number acting on a word yields a refined word (still **word**).

A number may also set a numeric dial of  $T$  — for example the rotation angle  $\alpha(r)$  — because that dial is already a scalar. That is how “wind of 12 m/s” can strengthen a turn without becoming a synonym of wind.

**Cast.** Rare and named. The map  $E_{\text{num} \rightarrow \text{word}}(2) = \text{two}$  turns a digit into a spoken numeral; the inverse is parsing. Casts are dictionary maps, not  $\oplus$ .

The illegal meeting is an untyped blend

$$\text{dawn} \oplus 37.4, \tag{24}$$

which has no meaning until one side is first cast into the other’s sort.

## 5.4 Operator legality

Table 1 records which buttons accept which pairs. Here  $+$  is schoolbook addition on the numeric encoding,  $\oplus$  is idempotent blend,  $T$  is typed action, and  $x - a + b$  is the analogical offset of Section 4.

| Pair                          | $+$ | $\oplus$ | $T_w(x)$ | $x - a + b$ |
|-------------------------------|-----|----------|----------|-------------|
| ( <b>word</b> , <b>word</b> ) | no  | yes      | yes      | yes         |
| ( <b>num</b> , <b>num</b> )   | yes | no       | no*      | no*         |
| ( <b>word</b> , <b>num</b> )  | no  | no       | typed    | no          |
| ( <b>num</b> , <b>word</b> )  | no  | no       | typed    | no          |

Table 1: Sort discipline on a single sphere. \*Numeric  $T$  and numeric offset may be defined later as maps of the chosen numeric encoding; they are not the lexical operators of Sections 3 and 4.

## 5.5 Worked mixed sentence

Take the report line *Wind of 12 m/s darkened the sun.* The register is used as follows.

1. Load wind, sort **word**.
2. Load 12, sort **num**. Juxtapose; do not form  $\text{wind} \oplus 12$ .

3. Load m/s as a unit-name (lexical qualifier of the number).
4. Load sun, sort **word**.
5. Let the number cap a strength,  $\sigma(12) \in [0, 1]$ , and set  $r_{\text{wind}} = \sigma(12)$  so that  $\alpha = \pi\sigma(12)$  in Equation (10).
6. Apply  $T_{\text{wind}}(\text{sun})$ . Output sort: **word**.

The scalar never became a hue. It turned a dial of an action whose actors remained words. That is the same division of labor a printed paper already uses: digits set magnitudes; words name what the magnitudes belong to.

## 5.6 Several lexicons, one concept sphere

Surface tokens are not meanings. Two dictionaries may write the same pair of concepts with unrelated strings:

$$\begin{aligned} L_1 : \quad & \mathbf{xyz} \mapsto \text{man}, \quad \mathbf{qyz} \mapsto \text{woman}, \\ L_2 : \quad & \mathbf{fgh} \mapsto \text{man}, \quad \mathbf{hgf} \mapsto \text{woman}. \end{aligned} \tag{25}$$

In the usual pipeline each lexicon is tokenized, encoded, and *embedded separately*. The geometry that makes man near king is then learned twice, and a further alignment step is required before  $L_1$  and  $L_2$  can share an analogy. The token-to-embedding map is doing two jobs at once: naming a string, and inventing a place for its meaning.

The unit sphere is offered as a *proposed common coordinate register*, not as a demonstrated universal semantic space and not as a replacement for a lexicon table. A token is a surface form. A state on  $\mathcal{S}$  is a meaning. Each lexicon is only a cast

$$E_L(\text{token}) = x \in \mathcal{S}, \tag{26}$$

the same kind of map as  $E_{\text{num} \rightarrow \text{word}}$  above. Under that discipline both **xyz** and **fgh** load the *same* state man. The offset king – man of Section 4 is then computed once, on the globe, and reused by every lexicon that points at those two points.

Embeddings therefore stay. They do not disappear. What changes is that they are no longer private vector spaces, one per lexicon. They are embeddings *into the same unit*  $\mathcal{S}$ . Once  $E_{L_1}(\mathbf{xyz}) = E_{L_2}(\mathbf{fgh}) = \text{man}$ , the two surface forms occupy one address, and every angle measured from that address is the same angle:

$$\angle(\text{man}, \text{woman}) = \angle(E_{L_1}(\mathbf{xyz}), E_{L_1}(\mathbf{qyz})) = \angle(E_{L_2}(\mathbf{fgh}), E_{L_2}(\mathbf{hgf})). \tag{27}$$

Under that proposal the embeddings would be uniform: same globe, same points for the same concepts, same angles for the same relations. A raw string still needs the lexicon map  $E_L$ . It would no longer need a new geometry. The algebra does not mint a universal man point by itself. That is a dictionary programme, not an axiom.

## 5.7 Deferred programme: standardized concept dictionaries

The present section only *names* the shared globe. It does not build the dictionaries, and it does not treat them as part of the algebra closed in Axioms 1–6. The later refinement, if the idea is pursued, is a pair of catalogues rather than a new operator, and is filed under Future Work (Section 10):

- one *concept dictionary* — a stipulated inventory of unit states on  $\mathcal{S}$  (or a higher-dimensional analogue), so that man and woman have standard addresses and therefore standard angles;
- one *language lexicon*  $E_L$  per written or spoken system, sending that language’s tokens onto those addresses;
- learnable paths — the species of  $T$  and the dials inside a given  $T$  (angle map, phase map, paint-versus-poem gate) may be fitted from examples. Learning would adjust the path, not the rule that a bowl is collected before it is read. See also Section 10.

Standardization of the first catalogue is what makes embeddings uniform across English, a cipher, or a second natural language. The second catalogue is still a tokenizer-plus-encoding: it is the phone book, not the city. Tokens and encodings have already earned their keep, so they stay. They only have to work *lighter*: look up a standard address instead of inventing one. They disappear as *inventors of geometry*. They do not disappear as *names*. A world in which every language pointed at the same concept dictionary would still print letters on a page; It would no longer need each corpus to learn a private man–woman angle. That programme is recorded here and left open. It is not an axiom, and it is not implied by the existence of  $\mathcal{S}$  alone.

## 5.8 Scope of the register model

This section claims only that one sphere can *host* both alphabets as a workspace, and that several lexicons can point at the same workspace through different casts. It does not claim that the two-dimensional surface  $S^2$  is a realistic embedding space for a full natural-language lexicon: a working word embedding typically needs tens or hundreds of dimensions. The construction here is a typed register and a toy dictionary on one globe. Scaling the lexical encoding is a later choice — a higher-dimensional sphere, a spherical-harmonic field, or an external lookup table that writes states into the register. The lookup tables  $E_L$  may still be learned; the claim is only that they should write into one geometry rather than each minting a private one. The unit ball of radius 1 in  $\mathbb{R}^3$  is where the idea is taught. The unit sphere in the embedding dimension is where close semantics can live. The algebra — collapse, path, result — is meant to travel with the dimension. It is not meant to replace 768 numbers with two angles and hope the library still sorts.

## 6 A Sketch of Conventions

These notes are a seed for later grammars, not a code to enforce. They show one way a typed register *might* be spoken. A later grammar — learned, cultural, or formal — may replace every line. This paper only needs the ground level enough that such a grammar has somewhere to stand.

**Case does not move the address.** The tokens *man*, *Man*, and *MAN* load the same direction  $(\theta, \phi)$ . Orthography is a doorbell, not a new city block.

**Language does not move the address.** English `man` and a Polish token for the same concept load the same point. Only the lexicon map  $E_L$  changes. Language is a phone-book tag  $L$ , not a second angle.

**Color is reserved for mood, not for language.** Phase  $\chi$  is already used as affect in the poem example (Section 3.5). If language or case also occupied  $\chi$ , that example would break. Emphasis may tint  $\chi$  or raise  $r$ ; it should not relocate `man` to `king`.

**A bag is not a sentence.**  $\{\text{sun, wind}\}$  is  $\oplus$ . “Wind darkens the sun” is  $T$ . Ingredients and order are different verbs. How often a unit is used is not a second shell in the bag: the path may visit `twinkle` twice; `num` may count  $1 + 1 + 1 + 1 = 4$ .  $\oplus$  will not.

**A numeral is not a word until it is cast.** The digit 2 stays `num`. The word `two` is `word`. There is no silent  $\oplus$  between them (Section 5).

A toy row makes the first three conventions visible. Place `man` at  $(\theta, \phi) = (10^\circ, 40^\circ)$ :

| Token            | $L$     | $(\theta, \phi)$       | $\chi$ (mood only)              |
|------------------|---------|------------------------|---------------------------------|
| <code>man</code> | English | $(10^\circ, 40^\circ)$ | $0^\circ$ (neutral)             |
| <code>Man</code> | English | $(10^\circ, 40^\circ)$ | $0^\circ$                       |
| <code>MAN</code> | English | $(10^\circ, 40^\circ)$ | $20^\circ$ (emphasis, optional) |
| <code>man</code> | Polish  | $(10^\circ, 40^\circ)$ | $0^\circ$                       |

Same block. Different doorbells. Nothing here is binding.

## 7 Operational Algebra & Numerical Demonstration

### 7.1 Vector Addition & Phase Blending

Collecting  $r = 0.6$  at  $0^\circ$  with  $r = 0.8$  at  $120^\circ$  (same place) yields two shells, not an angle. Applying  $\text{Mix} = T_{\text{paint}}$  and Equation (8) yields the 4-tuple  $(r', \theta', \phi', \chi') \approx (0.721, \theta, \phi, 73.9^\circ)$ . The older figure  $66.6^\circ$  is withdrawn: Axiom 5 as a length-only formula never produced it.

### 7.2 Rotational transposition, and what division is not

Rotation is Axiom 2.7:  $\mathbf{R} \in \text{SO}(3)$  turns  $\hat{n}$ . Inverse rotation is well-defined. Division as “deconvolution and signal unmixing” is not defined in this paper and is left open.

### 7.3 Spectral storage (optional, later)

Spherical-harmonic arrays of size  $O(L^2)$  may later store a field of many atoms. They are not the working atom of Axiom 2.2, and  $O(N^3) \rightarrow O(L^2)$  is not a theorem of Pattern Arithmetic as it stands.

## 8 Higher-order systems as bowls and paths

Higher-order truths are bowls of basic units read by a path. A storm, a forest, or a cancellation cartoon is such a reading. It is an example of the spine, not a model of the sky, the woods, or the clinic. The path defines the higher-order system: the bowl only lists who is present.

### 8.1 Atmospheric storm: one pipeline with numbers

This is a costume for the spine, not a model of the sky. Four incoming states, one bowl, two paths, two endpoints. Every figure below is Rodrigues' rotation of  $\hat{n}$  by  $\alpha = \pi r$  about the actor's axis, with  $\chi$  added. A reader can recompute.

#### 1. Input states.

$$\begin{aligned} \text{Wind} : r = 1, \hat{n} &= (1, 0, 0), \chi = 0^\circ, \tau = \text{word}, \quad \alpha = 180^\circ, \\ \text{Heat} : r = \frac{1}{2}, \hat{n} &= (0, 1, 0), \chi = 90^\circ, \tau = \text{word}, \quad \alpha = 90^\circ, \\ \text{Moisture} : r = \frac{2}{3}, \hat{n} &= (0, 0, 1), \chi = 45^\circ, \tau = \text{word}, \quad \alpha = 120^\circ, \\ \text{Eye} : r = 1, \hat{n} &= \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right), \chi = 0^\circ, \tau = \text{word}. \end{aligned}$$

The eye sits at  $(\theta, \phi) = (45^\circ, 0^\circ)$ .

#### 2. Collapse into the bowl.

$$R = \text{Wind} \oplus \text{Heat} \oplus \text{Moisture} \oplus \text{Eye} = \{\text{Wind}, \text{Heat}, \text{Moisture}, \text{Eye}\}. \quad (28)$$

Four distinct names, four seats. A second Wind would not open a fifth.  $R$  is a bowl, not a blended 4-tuple. Mix is a different reading and is not the storm.

**3. Two trajectory paths.** The register occupant is the eye. The other three shells act, in order:

$$T_{\text{deepen}} = T_{\text{Wind}} \circ T_{\text{Heat}} \circ T_{\text{Moisture}}, \quad T_{\text{clear}} = T_{\text{Moisture}} \circ T_{\text{Heat}} \circ T_{\text{Wind}}. \quad (29)$$

*Deepen* — moisture, then heat, then wind:

$$\begin{aligned} T_{\text{Moisture}}(\text{Eye}) &= (1, (-\frac{\sqrt{2}}{4}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{2}), 45^\circ), \\ T_{\text{Heat}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{4}), 135^\circ), \\ T_{\text{Wind}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4}), 135^\circ). \end{aligned}$$

*Clear* — wind, then heat, then moisture:

$$\begin{aligned} T_{\text{Wind}}(\text{Eye}) &= (1, (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 0^\circ), \\ T_{\text{Heat}}(\cdots) &= (1, (-\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 90^\circ), \\ T_{\text{Moisture}}(\cdots) &= (1, (\frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2}), 135^\circ). \end{aligned}$$

**4. Endpoints.**  $T$  leaves the occupant’s strength at  $r = 1$ . Phase addition commutes, so both paths end at  $\chi = 135^\circ$ . The directions do not agree:

$$T_{\text{deepen}}(\text{Eye}) : \quad \hat{n} = \left( \frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4} \right) \approx (0.7071, -0.6124, -0.3536), \\ (\theta, \phi) \approx (110.7^\circ, 319.1^\circ);$$

$$T_{\text{clear}}(\text{Eye}) : \quad \hat{n} = \left( \frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2} \right) \approx (0.3536, -0.6124, -0.7071), \\ (\theta, \phi) = (135^\circ, 300^\circ).$$

Same bowl, two weathers. The storm is the walk. It is not a fifth pin, and it is not a claim about vorticity or shear.

## 8.2 Forest (costume)

Flora, fauna, soil, moisture sit as units.  $\oplus$  gathers them. There is no  $A + B = AB$ . A path may read competition or succession. The forest-as-system is that reading.

## 8.3 Pathogen cartoon (costume)

A mark and an equal-strength anti-phase mark may be *read* by Mix into black. That is geometry. It is not medicine.

# 9 Discussion & Related Work

Each paragraph below states what is reused and what is added. None of these citations is a claim of identity. Empty headings are omitted.

**Bloch sphere.** A pure qubit is a point on  $S^2$ ; a mixed qubit lives in the ball. Pattern Arithmetic reuses that unit globe. It differs in three respects. Strength  $r \leq 1$  is an activation of one arrow, not a purity parameter of a density matrix. Phase  $\chi$  is an internal colour of that arrow, not a global  $U(1)$  that can be quotiented away. There is no measurement postulate:  $\oplus$  and Mix mint a third unit or black; they do not collapse a superposition onto an eigenstate. Related work may say: we reuse a unit globe; we do not borrow quantum authority for  $\oplus$ .

**Word2vec and cosine embeddings.** Lexical similarity as an angle is reused. A bag of words as a commutative mix is reused in spirit. What is added is a bounded unit ( $r \leq 1$ ), an explicit collection operator that is idempotent ( $A \oplus A = A$ ), and an order-dependent path  $T$  that cosine addition does not supply. Offset is a manifold cousin of the familiar king – man + woman arithmetic, not a proof that the toy globe is a full lexicon.

**RotatE.** Knowledge-graph embeddings that treat a relation as a rotation in a complex plane are the nearest neighbour of  $T$ . Pattern Arithmetic extends that idea from  $S^1$  (or a product of circles) to  $SO(3)$  acting on  $S^2$ , and pairs the rotation with an additive phase  $\chi$ . It also keeps a separate collapse operator, which RotatE does not.

**von Mises–Fisher.** Directional statistics on the sphere reuse the same stage. Those models estimate a density. Pattern Arithmetic does not: the atom is a point,  $r$  is not a concentration  $\kappa$ , and  $\oplus$ ,  $T$ , and offset are algebraic operations on units, not maximum-likelihood fits.

**Riemann sphere.** Stereographic compactification of  $\mathbb{C} \cup \{\infty\}$  onto  $S^2$  is the natural home of one numeric encoding (unbounded reals toward the poles). The algebra of this paper is not the Möbius group. A clock, or a stereo map, is a chart;  $+$  in `num` mode is decoded arithmetic, not a fractional linear map.

**SO(3).** The transformation  $T$  is a named action of the rotation group, plus a phase shift. That is the whole geometric engine of a path. What SO(3) does not supply is the bowl, the cap  $r \leq 1$ , the sort  $\tau$ , or Mix. Non-commutativity of poems is inherited; idempotence of  $\oplus$  is not.

**Spherical harmonics.** A later field representation of many arrows could be stored as  $Y_\ell^m$  coefficients, replacing an  $O(N^3)$  voxel cube by an  $O(L^2)$  spectral array. That is a compression option for a field, not a definition of the atom, and it is not claimed as a result here.

## 10 Conclusion & Future Directions

This paper has sketched a single pipeline: collapse many sorted units into a bowl of typed atoms  $x = (r, \hat{n}, \chi, \tau)$  on a unit sphere, then derive results by ordered transformation of a register occupant. Embeddings, on this view, are not abolished. They are embeddings into one proposed common coordinate register, so that a concept such as man *may* keep the same angles under every lexicon that points at it. That is a design, not a demonstrated universal semantic space.

A few spoken conventions are sketched in Section 6 as seed, not statute.

### 10.1 Future work

What remains to be built, and is only filed here, is a standardized concept dictionary together with per-language lexica  $E_L$  (Section 5.7). Until those catalogues exist, tokenization and encoding remain the handles by which a paragraph finds its points. They may continue; they will only have to work lighter. The open bet is that they need not also remain the architects of a new space of meanings. The algebra closed in this paper does not produce a universal man point by itself. Standardization is sociological and empirical; it is not an axiom.

A further deferred question is whether a path  $T$  should stay hand-specified or carry learnable weights: the angle law  $\alpha(r)$ , a phase map, a gate between  $T_{\text{paint}}$  and  $T_{\text{poem}}$ , and the lexicon casts  $E_L$ . The honest constraint is structural. If undo-the-poem matters, learning should stay inside SO(3), not collapse  $T$  into an arbitrary matrix. Weights may change how the shells are thrown. They should not let the shells throw themselves.

Ambient  $\mathbb{R}^3$  makes the expressions  $\mathcal{S} + (a, b, c)$  and  $\mathcal{S} \cdot (a, b, c)$  look legal. They are legal as *vector* arithmetic. They are not automatically legal as *unit* arithmetic. A state with  $r = 0.98$  added to another with  $r = 0.98$  along the same direction is  $r = 1.96$ ,

which has left the ball. A scalar  $2 \cdot (0.98)$  does the same. The unit cap  $r \in [0, 1]$  then forces a choice, to be made later and not here: clip (overflow saturates at 1), project (renormalize direction and forget extra strength), or refuse (the operator is undefined off the ball). Schoolbook  $+$  on the `num` encoding is a different door: decode, add in  $\mathbb{R}$ , encode back if the chart allows. Do not let  $\mathbb{R}^3$  addition quietly replace  $\oplus$ , Mix, or  $T$ . The embedding lives in  $\mathbb{R}^3$ . The unit lives in the ball. As already noted in Section 5.8, the three-dimensional ball teaches the algebra; close document semantics, if pursued, live on a higher-dimensional sphere that carries the same verbs.